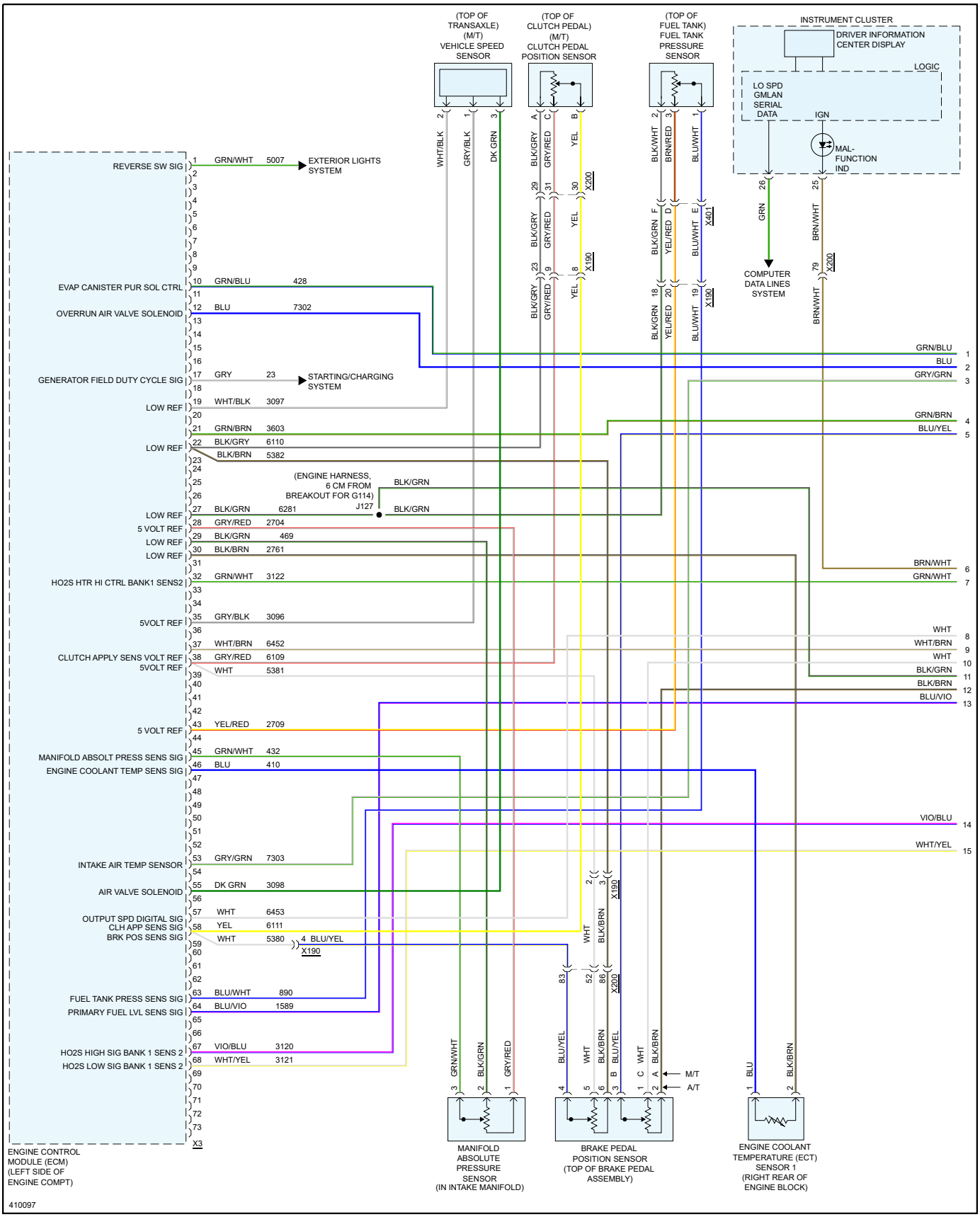
2013 Chevrolet Cruze L4-1.8L

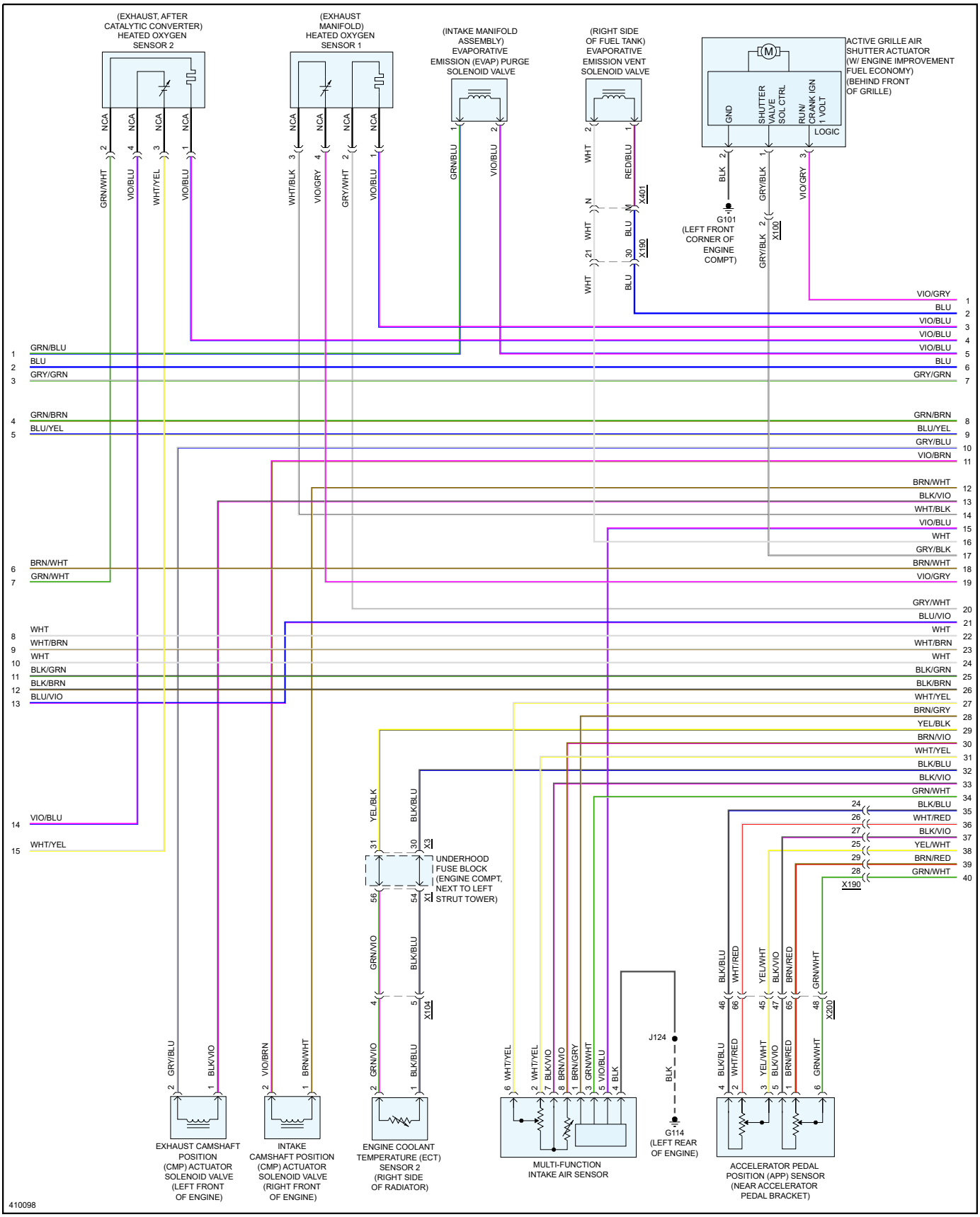
Vehicle > Powertrain Management > Diagrams > Electrical - Interactive Color (Non OE)

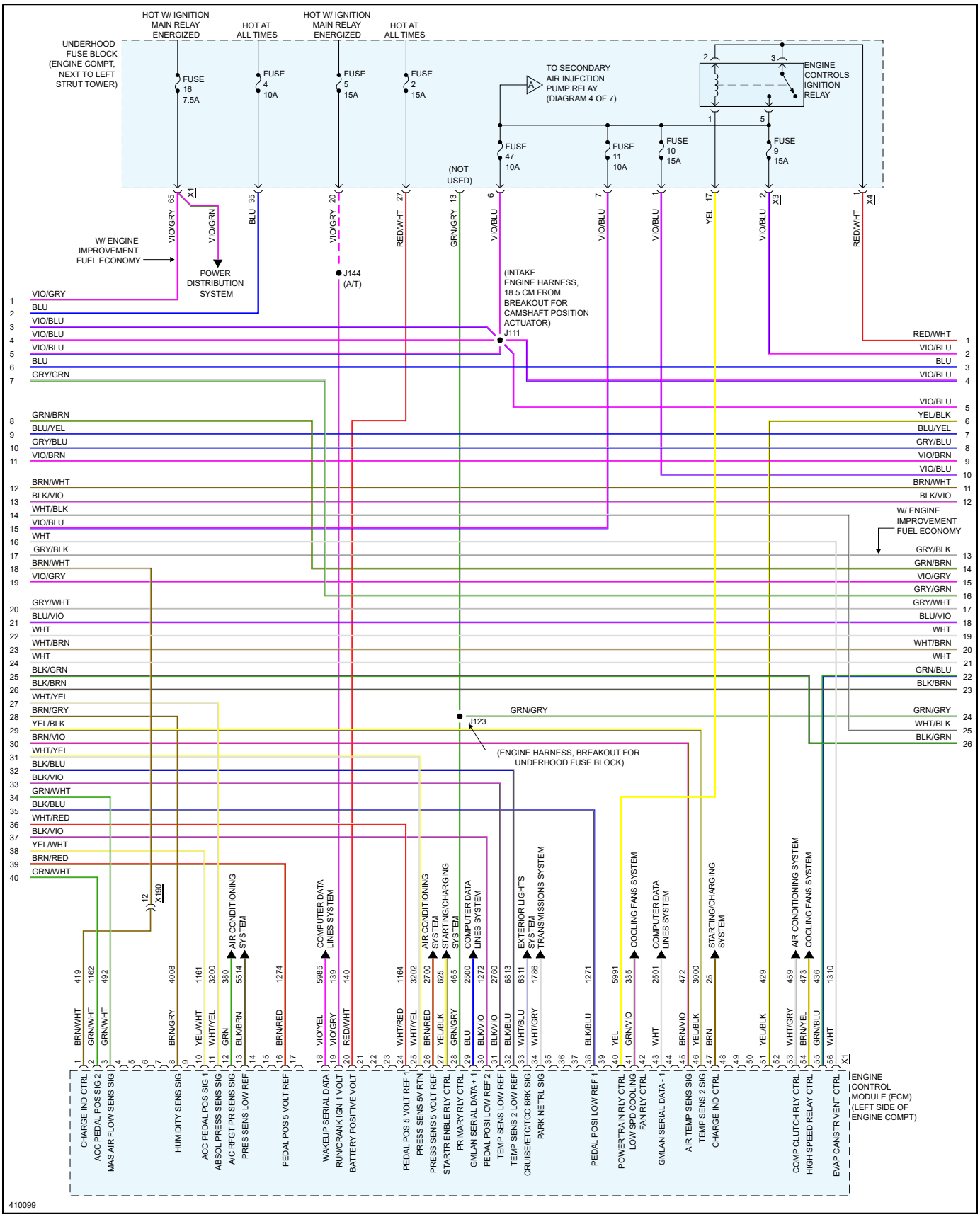
ENGINE CONTROLS

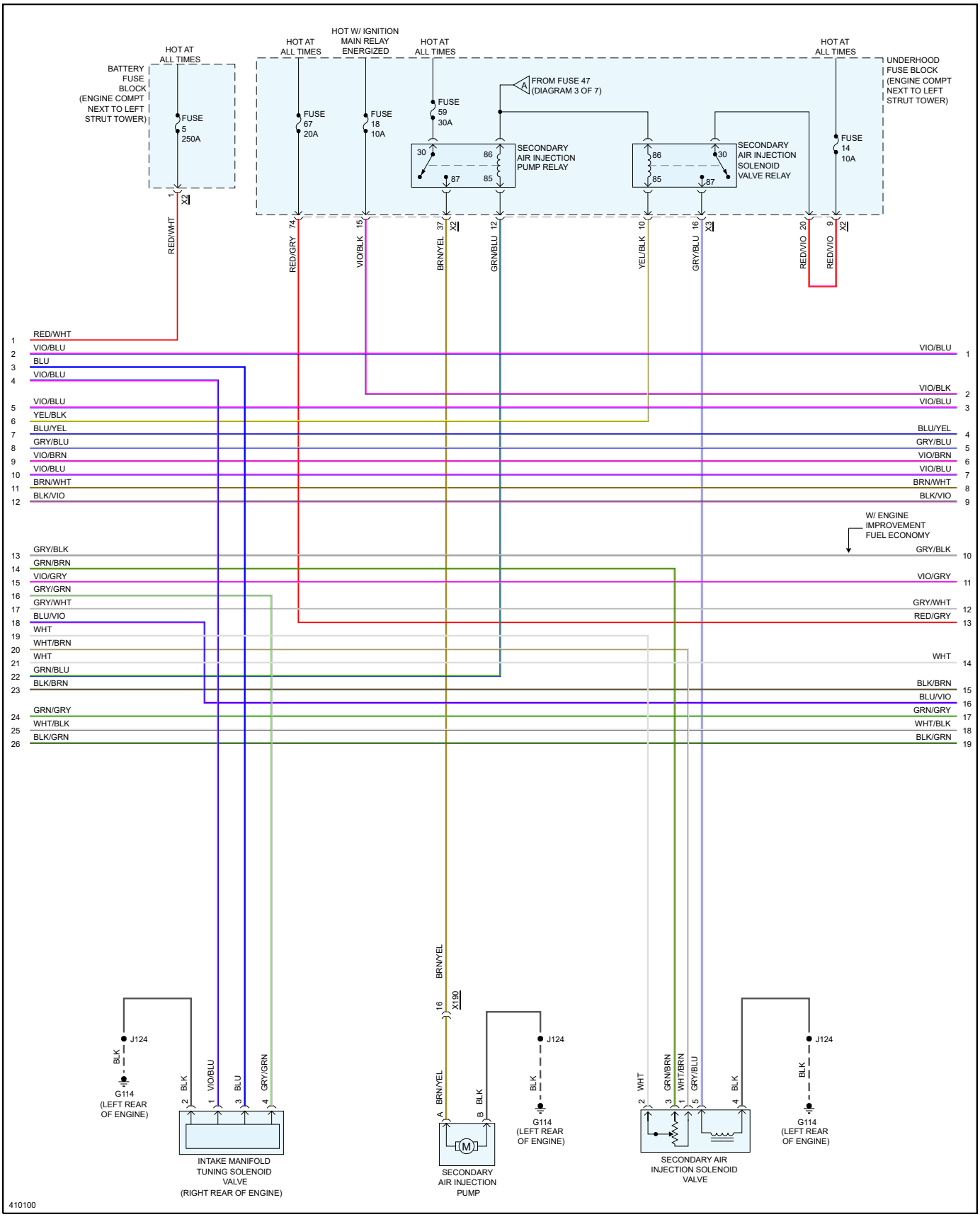
Engine Controls

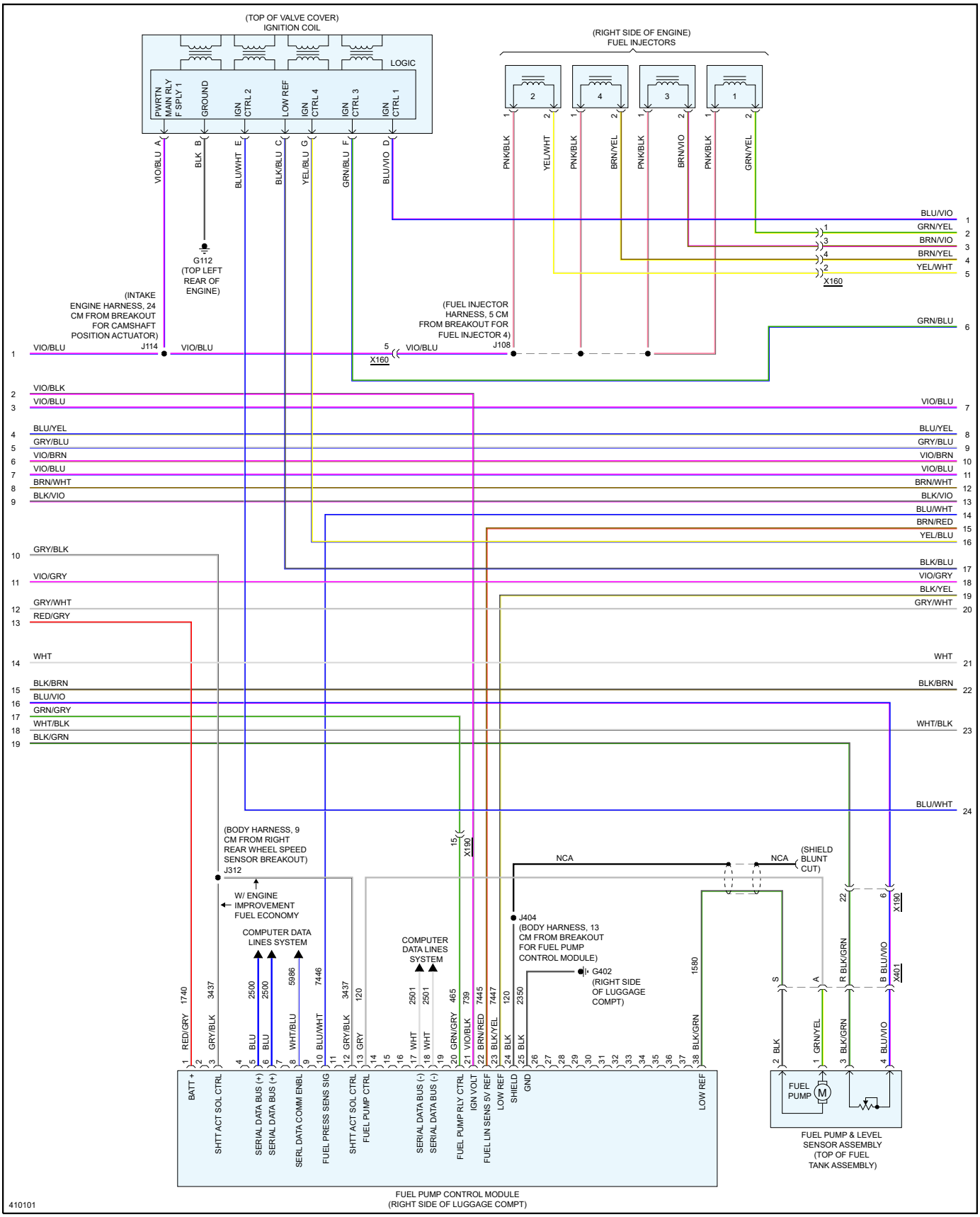
Page 1 of 7

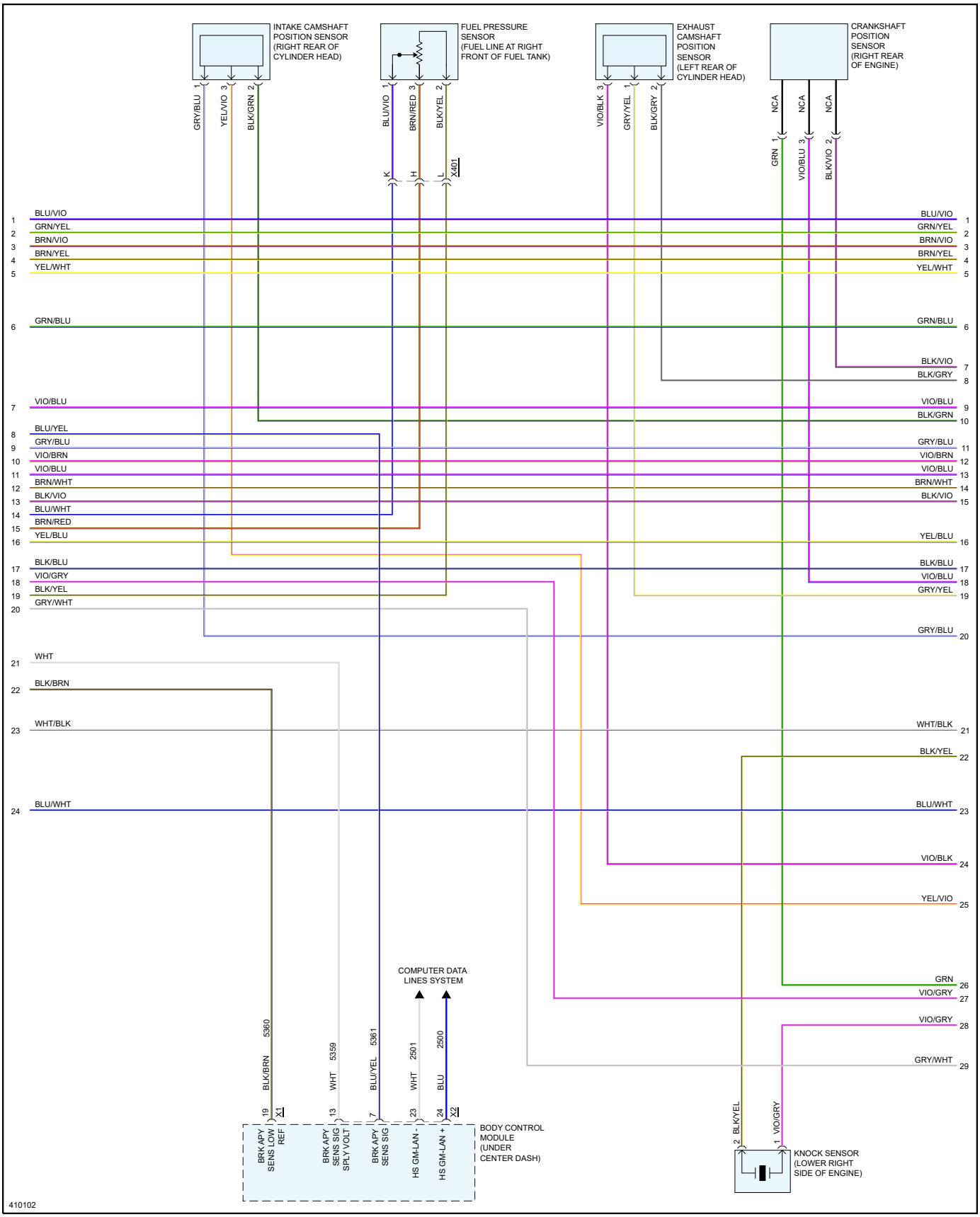

(TOP OF TRANSAXLE) (M/T) VEHICLE SPEED SENSOR
(TOP OF CLUTCH PEDAL) (M/T) CLUTCH PEDAL POSITION SENSOR
(TOP OF FUEL TANK) FUEL TANK PRESSURE SENSOR
INSTRUMENT CLUSTER
DRIVER INFORMATION CENTER DISPLAY
LOGIC
LO SPD GMLAN SERIAL DATA
IGN
MAL-FUNCTION IND
REVERSE SW SIG
EXTERIOR LIGHTS SYSTEM
EVAP CANISTER PUR SOL CTRL
OVERRUN AIR VALVE SOLENOID
GENERATOR FIELD DUTY CYCLE SIG
STARTING/CHARGING SYSTEM
LOW REF
5 VOLT REF
HO2S HTR HI CTRL BANK1 SENS2
5VOLT REF
CLUTCH APPLY SENS VOLT REF
INTAKE AIR TEMP SENSOR
AIR VALVE SOLENOID
OUTPUT SPD DIGITAL SIG
CLH APP SENS SIG
BRK POS SENS SIG
FUEL TANK PRESS SENS SIG
PRIMARY FUEL LVL SENS SIG
MANIFOLD ABSOLUTE PRESSURE SENSOR
HO2S HIGH SIG BANK 1 SENS 2
HO2S LOW SIG BANK 1 SENS 2
(ENGINE HARNESS, 6 CM FROM BREAKOUT FOR G114)
J127
COMPUTER DATA LINES SYSTEM
M/T
A/T
BRAKE PEDAL POSITION SENSOR (TOP OF BRAKE PEDAL ASSEMBLY)
ENGINE COOLANT TEMPERATURE (ECT) SENSOR 1 (RIGHT REAR OF ENGINE BLOCK)
ENGINE CONTROL MODULE (ECM) (LEFT SIDE OF ENGINE COMPT)
X3
X190
X200
X401
410097

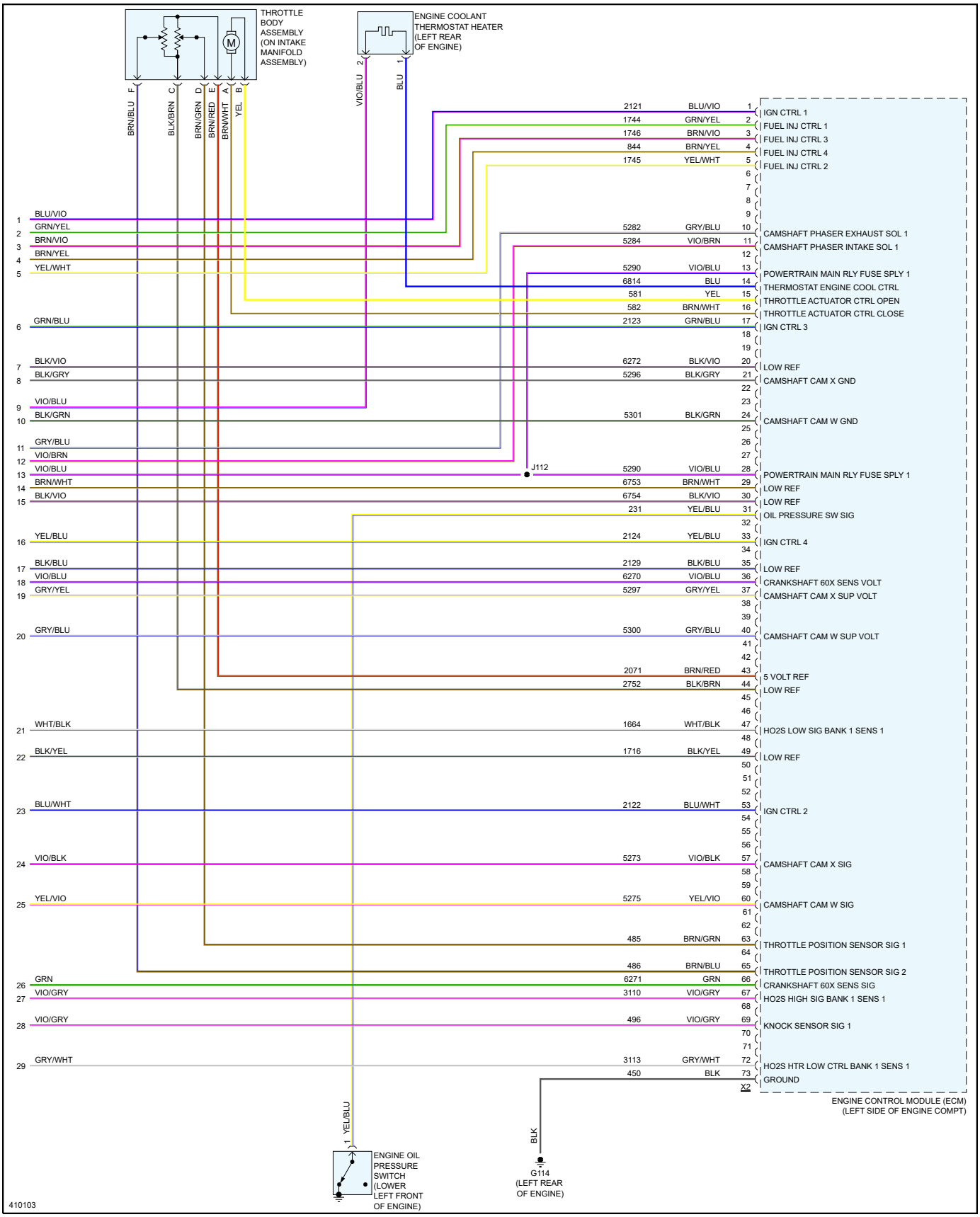

(EXHAUST, AFTER CATALYTIC CONVERTER) HEATED OXYGEN SENSOR 2
(EXHAUST MANIFOLD) HEATED OXYGEN SENSOR 1
(INTAKE MANIFOLD ASSEMBLY) EVAPORATIVE EMISSION (EVAP) PURGE SOLENOID VALVE
(RIGHT SIDE OF FUEL TANK) EVAPORATIVE EMISSION VENT SOLENOID VALVE
ACTIVE GRILLE AIR SHUTTER ACTUATOR (W/ ENGINE IMPROVEMENT FUEL ECONOMY) (BEHIND FRONT OF GRILLE)
GND
SHUTTER VALVE SOL CTRL
RUN/ CRANK IGN 1 VOLT
LOGIC
G101 (LEFT FRONT CORNER OF ENGINE COMPT)
G114 (LEFT REAR OF ENGINE)
J124
X100
X190
X401
X3
X1
X104
X200
UNDERHOOD FUSE BLOCK (ENGINE COMPT, NEXT TO LEFT STRUT TOWER)
EXHAUST CAMSHAFT POSITION (CMP) ACTUATOR SOLENOID VALVE (LEFT FRONT OF ENGINE)
INTAKE CAMSHAFT POSITION (CMP) ACTUATOR SOLENOID VALVE (RIGHT FRONT OF ENGINE)
ENGINE COOLANT TEMPERATURE (ECT) SENSOR 2 (RIGHT SIDE OF RADIATOR)
MULTI-FUNCTION INTAKE AIR SENSOR
ACCELERATOR PEDAL POSITION (APP) SENSOR (NEAR ACCELERATOR PEDAL BRACKET)
410098


HOT W/ IGNITION MAIN RELAY ENERGIZED
HOT AT ALL TIMES
HOT W/ IGNITION MAIN RELAY ENERGIZED
HOT AT ALL TIMES
UNDERHOOD FUSE BLOCK (ENGINE COMPT, NEXT TO LEFT STRUT TOWER)
FUSE 16 7.5A
FUSE 4 10A
FUSE 5 15A
FUSE 2 15A
TO SECONDARY AIR INJECTION PUMP RELAY (DIAGRAM 4 OF 7)
ENGINE CONTROLS IGNITION RELAY
FUSE 47 10A
FUSE 11 10A
FUSE 10 15A
FUSE 9 15A
(NOT USED)
W/ ENGINE IMPROVEMENT FUEL ECONOMY
POWER DISTRIBUTION SYSTEM
J144 (A/T)
(INTAKE ENGINE HARNESS, 18.5 CM FROM BREAKOUT FOR CAMSHAFT POSITION ACTUATOR)
J111
J123
(ENGINE HARNESS, BREAKOUT FOR UNDERHOOD FUSE BLOCK)
X190
ENGINE CONTROL MODULE (ECM) (LEFT SIDE OF ENGINE COMPT)
410099


HOT AT ALL TIMES
BATTERY FUSE BLOCK (ENGINE COMPT NEXT TO LEFT STRUT TOWER)
FUSE 5 250A
HOT AT ALL TIMES
HOT W/ IGNITION MAIN RELAY ENERGIZED
HOT AT ALL TIMES
HOT AT ALL TIMES
UNDERHOOD FUSE BLOCK (ENGINE COMPT NEXT TO LEFT STRUT TOWER)
FUSE 67 20A
FUSE 18 10A
FUSE 59 30A
A
FROM FUSE 47 (DIAGRAM 3 OF 7)
SECONDARY AIR INJECTION PUMP RELAY
SECONDARY AIR INJECTION SOLENOID VALVE RELAY
FUSE 14 10A
RED/WHT
RED/GRY
VIO/BLK
BRN/YEL
GRN/BLU
YEL/BLK
GRY/BLU
RED/VIO
RED/VIO
W/ ENGINE IMPROVEMENT FUEL ECONOMY
J124
G114 (LEFT REAR OF ENGINE)
INTAKE MANIFOLD TUNING SOLENOID VALVE (RIGHT REAR OF ENGINE)
SECONDARY AIR INJECTION PUMP
SECONDARY AIR INJECTION SOLENOID VALVE
410100


(TOP OF VALVE COVER)
IGNITION COIL
LOGIC
PWRTN MAIN RLY F SPLY 1
GROUND
IGN CTRL 2
LOW REF
IGN CTRL 4
IGN CTRL 3
IGN CTRL 1
(RIGHT SIDE OF ENGINE)
FUEL INJECTORS
G112
(TOP LEFT REAR OF ENGINE)
(INTAKE ENGINE HARNESS, 24 CM FROM BREAKOUT FOR CAMSHAFT POSITION ACTUATOR)
J114
(FUEL INJECTOR HARNESS, 5 CM FROM BREAKOUT FOR FUEL INJECTOR 4)
J108
X160
(BODY HARNESS, 9 CM FROM RIGHT REAR WHEEL SPEED SENSOR BREAKOUT)
J312
W/ ENGINE IMPROVEMENT FUEL ECONOMY
COMPUTER DATA LINES SYSTEM
J404
(BODY HARNESS, 13 CM FROM BREAKOUT FOR FUEL PUMP CONTROL MODULE)
G402
(RIGHT SIDE OF LUGGAGE COMPT)
NCA
(SHIELD BLUNT CUT)
X190
X401
FUEL PUMP
FUEL PUMP & LEVEL SENSOR ASSEMBLY
(TOP OF FUEL TANK ASSEMBLY)
FUEL PUMP CONTROL MODULE
(RIGHT SIDE OF LUGGAGE COMPT)
410101


INTAKE CAMSHAFT POSITION SENSOR (RIGHT REAR OF CYLINDER HEAD)
FUEL PRESSURE SENSOR (FUEL LINE AT RIGHT FRONT OF FUEL TANK)
EXHAUST CAMSHAFT POSITION SENSOR (LEFT REAR OF CYLINDER HEAD)
CRANKSHAFT POSITION SENSOR (RIGHT REAR OF ENGINE)
GRY/BLU
YEL/VIO
BLK/GRN
BLU/VIO
BRN/RED
BLK/YEL
VIO/BLK
GRY/YEL
BLK/GRY
NCA
GRN
VIO/BLU
BLK/VIO
X401
BLU/VIO
GRN/YEL
BRN/VIO
BRN/YEL
YEL/WHT
GRN/BLU
BLK/VIO
BLK/GRY
VIO/BLU
BLK/GRN
BLU/YEL
GRY/BLU
VIO/BRN
VIO/BLU
BRN/WHT
BLK/VIO
BLU/WHT
BRN/RED
YEL/BLU
BLK/BLU
VIO/GRY
BLK/YEL
GRY/WHT
WHT
BLK/BRN
WHT/BLK
BLU/WHT
VIO/BLK
YEL/VIO
GRN
VIO/GRY
GRY/WHT
COMPUTER DATA LINES SYSTEM
BLK/BRN
WHT
BLU/YEL
WHT
BLU
5360
5359
5361
2501
2500
X1
X2
BRK APY SENS LOW REF
BRK APY SENS SIG
SPLY VOLT
BRK APY SENS SIG
HS GM-LAN -
HS GM-LAN +
BODY CONTROL MODULE (UNDER CENTER DASH)
BLK/YEL
VIO/GRY
KNOCK SENSOR (LOWER RIGHT SIDE OF ENGINE)
410102


THROTTLE BODY ASSEMBLY (ON INTAKE MANIFOLD ASSEMBLY)
ENGINE COOLANT THERMOSTAT HEATER (LEFT REAR OF ENGINE)
BRN/BLU F
BLK/BRN C
BRN/GRN D
BRN/RED E
BRN/WHT A
YEL B
VIO/BLU 2
BLU 1
1 BLU/VIO
2 GRN/YEL
3 BRN/VIO
4 BRN/YEL
5 YEL/WHT
6 GRN/BLU
7 BLK/VIO
8 BLK/GRY
9 VIO/BLU
10 BLK/GRN
11 GRY/BLU
12 VIO/BRN
13 VIO/BLU
14 BRN/WHT
15 BLK/VIO
16 YEL/BLU
17 BLK/BLU
18 VIO/BLU
19 GRY/YEL
20 GRY/BLU
21 WHT/BLK
22 BLK/YEL
23 BLU/WHT
24 VIO/BLK
25 YEL/VIO
26 GRN
27 VIO/GRY
28 VIO/GRY
29 GRY/WHT
J112
2121 BLU/VIO 1 IGN CTRL 1
1744 GRN/YEL 2 FUEL INJ CTRL 1
1746 BRN/VIO 3 FUEL INJ CTRL 3
844 BRN/YEL 4 FUEL INJ CTRL 4
1745 YEL/WHT 5 FUEL INJ CTRL 2
6
7
8
9
5282 GRY/BLU 10 CAMSHAFT PHASER EXHAUST SOL 1
5284 VIO/BRN 11 CAMSHAFT PHASER INTAKE SOL 1
12
5290 VIO/BLU 13 POWERTRAIN MAIN RLY FUSE SPLY 1
6814 BLU 14 THERMOSTAT ENGINE COOL CTRL
581 YEL 15 THROTTLE ACTUATOR CTRL OPEN
582 BRN/WHT 16 THROTTLE ACTUATOR CTRL CLOSE
2123 GRN/BLU 17 IGN CTRL 3
18
19
6272 BLK/VIO 20 LOW REF
5296 BLK/GRY 21 CAMSHAFT CAM X GND
22
23
5301 BLK/GRN 24 CAMSHAFT CAM W GND
25
26
27
5290 VIO/BLU 28 POWERTRAIN MAIN RLY FUSE SPLY 1
6753 BRN/WHT 29 LOW REF
6754 BLK/VIO 30 LOW REF
231 YEL/BLU 31 OIL PRESSURE SW SIG
32
2124 YEL/BLU 33 IGN CTRL 4
34
2129 BLK/BLU 35 LOW REF
6270 VIO/BLU 36 CRANKSHAFT 60X SENS VOLT
5297 GRY/YEL 37 CAMSHAFT CAM X SUP VOLT
38
39
5300 GRY/BLU 40 CAMSHAFT CAM W SUP VOLT
41
42
2071 BRN/RED 43 5 VOLT REF
2752 BLK/BRN 44 LOW REF
45
46
1664 WHT/BLK 47 HO2S LOW SIG BANK 1 SENS 1
48
1716 BLK/YEL 49 LOW REF
50
51
52
2122 BLU/WHT 53 IGN CTRL 2
54
55
56
5273 VIO/BLK 57 CAMSHAFT CAM X SIG
58
59
5275 YEL/VIO 60 CAMSHAFT CAM W SIG
61
62
485 BRN/GRN 63 THROTTLE POSITION SENSOR SIG 1
64
486 BRN/BLU 65 THROTTLE POSITION SENSOR SIG 2
6271 GRN 66 CRANKSHAFT 60X SENS SIG
3110 VIO/GRY 67 HO2S HIGH SIG BANK 1 SENS 1
68
496 VIO/GRY 69 KNOCK SENSOR SIG 1
70
71
3113 GRY/WHT 72 HO2S HTR LOW CTRL BANK 1 SENS 1
450 BLK 73 GROUND
X2
ENGINE CONTROL MODULE (ECM) (LEFT SIDE OF ENGINE COMPT)
YEL/BLU
1
ENGINE OIL PRESSURE SWITCH (LOWER LEFT FRONT OF ENGINE)
BLK
G114 (LEFT REAR OF ENGINE)
410103